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# Health Anxiety, AI Chatbot Use and Cyberchondria Among Adults in Baghdad: An Online Cross-Sectional Survey

A graduation research project submitted to the College of Medicine, University of Baghdad, in partial fulfilment of the requirements for the degree of Bachelor of Medicine and Bachelor of Surgery (M.B.Ch.B.)

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# List of Abbreviations

| | |
|---|---|
| **AI** | Artificial intelligence |
| **AOR** | Adjusted odds ratio |
| **AUC** | Area under the receiver operating characteristic curve |
| **CI** | Confidence interval |
| **CSS-12** | Cyberchondria Severity Scale, 12-item short form |
| **ER** | Emergency room |
| **GP** | General practitioner |
| **HA** | Health anxiety |
| **IQR** | Interquartile range |
| **LLM** | Large language model |
| **OR** | Odds ratio |
| **SD** | Standard deviation |
| **SHAI-14** | Short Health Anxiety Inventory, 14-item version |
| **STROBE** | Strengthening the Reporting of Observational Studies in Epidemiology |
| **VIF** | Variance inflation factor |

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## Summary / Abstract

**Background:** Many adults look up symptoms online before seeing a doctor, increasingly with artificial-intelligence (AI) chatbots. Health anxiety and cyberchondria (excessive, distressing online health searching) are thought to feed each other, but evidence from Iraq, and on AI chatbots, is scarce.

**Objectives:** To measure the association between health anxiety and cyberchondria among adults in Baghdad, to compare medical students with other participants, and to describe online and AI health-information seeking and behaviour before consulting a doctor.

**Methods:** Cross-sectional online survey of 300 adults in Baghdad (July–September 2026) using the Short Health Anxiety Inventory (SHAI-14) and the Cyberchondria Severity Scale (CSS-12), analysed with non-parametric group comparisons, $\chi^2$ tests, correlation, and multiple linear and binary logistic regression.

**Results:** Participants were mostly young (62.3% aged 20–29). AI chatbots were ticked by 78.0% (88.7% including those who named a chatbot), more than search engines (56.3%). Health anxiety correlated with cyberchondria ($r = 0.47$, 95% CI 0.38–0.55) and remained its strongest predictor after adjustment ($\beta = 0.48$), most strongly with excessive searching and post-search distress. Elevated health anxiety (SHAI-14 ≥ 18, a screening threshold, not a diagnosis) was found in 19.0%. After searching, 34.7% delayed or skipped care because they felt reassured and 31.7% never told their doctor, citing fear of judgement. Medical students had higher health anxiety than other participants before adjustment (13.3 vs 11.6), but not after adjustment for age and other factors, and their cyberchondria was the same. The association between AI use and cyberchondria was not robust to how AI use was defined.

**Conclusion:** In this young, health-oriented convenience sample, health anxiety and cyberchondria were linked and AI chatbots were a major source of health information, without reliable evidence that they add to cyberchondria beyond health anxiety. Clinicians should ask patients, without judgement, what they have searched or asked an AI.

**Keywords:** cyberchondria; health anxiety; artificial intelligence; chatbots; Iraq.

# 1. Introduction

## 1.1 Background

Many people now look a symptom up online before they see a doctor. Often it goes into a search engine, or is described to an AI chatbot, well before a physician hears about it. The information is immediate, free and available at any hour, and it can help people decide whether a symptom needs medical attention. For a sizeable minority, however, searching makes the worry worse, a pattern known as cyberchondria [1,2]. About four in five Iraqis were online by 2025 [3], yet we found no study from Baghdad on how this habit interacts with health anxiety or affects the consultation that follows.

## 1.2 Health anxiety

Health anxiety is a persistent preoccupation with having, or getting, a serious illness, kept going by misreading normal bodily sensations. It runs on a continuum from passing worry to illness anxiety disorder and somatic symptom disorder in the DSM-5-TR [4]. In the cognitive-behavioural model, sensations are read as signs of serious disease, and the resulting anxiety is maintained by safety behaviours such as body checking and reassurance seeking, which relieve it briefly but stop the person learning that the feared outcome does not happen [5].

Health anxiety is common and costly: its lifetime prevalence is about 6% in the general population and up to 20% in hospital outpatients, and it generates unnecessary medical contacts [6]. In Iraq, 49.5% of 1591 respondents to an online survey during the COVID-19 restrictions screened positive for health anxiety on the Health Anxiety Questionnaire [7].

### 1.3 Cyberchondria

White and Horvitz showed in large-scale search logs that searches for common, harmless symptoms often escalated towards rare and serious diagnoses [1]. Starcevic and Berle defined cyberchondria as excessive or repeated online health searching that increases, rather than relieves, distress [2]; searching itself is not pathological, but becomes cyberchondria when it is excessive, compulsive and distressing. The 33-item Cyberchondria Severity Scale has a 12-item short form (CSS-12) that keeps four subscales (excessiveness, compulsion, distress and reassurance seeking) [8], and an Arabic version has shown acceptable structure in Lebanon [9].

### 1.4 Health anxiety and cyberchondria

The two are closely related but not identical: beyond its overlap with health anxiety, cyberchondria is linked with more functional impairment and healthcare use [10], and in a randomised experiment searching the causes of provoked bodily symptoms raised health anxiety and the wish to see a physician [11]. Two meta-analyses found that health anxiety correlates strongly with cyberchondria ($r = 0.62$ and 0.63), with subscale correlations from 0.24 to 0.66 [12,13]. Whether the two fuel each other is unsettled. A four-wave Dutch panel of 5322 adults found early evidence of a two-way relationship, though only in people with non-clinical health anxiety [14].

The terms used in this thesis overlap, so Box 1 sets out how they differ and how each was measured.

**Box 1.** Health anxiety, illness anxiety disorder, cyberchondria and ordinary online health searching

| | Health anxiety | Illness anxiety disorder | Cyberchondria | Ordinary online health searching |
|---|---|---|---|---|
| What it is | Worry about having or getting a serious illness, on a continuum from mild to severe | The DSM-5-TR diagnosis at the severe end: preoccupation for at least 6 months | Excessive or repeated online health searching that increases rather than relieves distress | Looking up information about a symptom, condition or treatment |
| Key features | Misreading bodily sensations; checking; reassurance seeking | No or mild somatic symptoms; high health anxiety; excessive checking or avoidance | Excessiveness, compulsion, distress and reassurance seeking | Occasional, goal-directed; usually reduces uncertainty |
| Status | Dimension (screened, not diagnosed) | Clinical diagnosis by a clinician | Behavioural pattern, not a diagnosis | Normal behaviour |
| In this study | SHAI-14 (0–42; ≥ 18 elevated) | Not assessed | CSS-12 (12–60; four subscales) | Tools, motives and behaviour items |

*Note.* Sources: DSM-5-TR [4]; cognitive model [5]; cyberchondria [2,8].

## 1.5 From search engines to AI chatbots

Most of this evidence predates large language model chatbots such as ChatGPT and Gemini, which answer in fluent, personal language, can be questioned again and again, and can seem more empathetic than physicians [15]. In a large US survey about one in five respondents had used ChatGPT for health information, and about a third of them changed medications or asked for referrals based on its answers [16]. It remains unclear whether chatbots relieve worry with calm answers or feed it by always being available for another round of reassurance.

## 1.6 Searching and the doctor–patient relationship

Online information can help the relationship when patients discuss it with their physician, and strain it when they hide it [17]. Reassurance seeking may lead to extra consultations, while false reassurance may delay care. In the Arab region, cyberchondria was common in Jeddah [18] and linked with health anxiety in Saudi primary care [19], and across 13 Arabic-speaking countries trust in online information predicted delayed consultation [20]. No comparable study exists from Iraq.

## 1.7 Rationale

This study addresses three gaps: it gives, to our knowledge, the first estimate of the association between health anxiety and cyberchondria (CSS-12) in Baghdad; it examines AI chatbots separately from search engines; and it asks whether medical students, who learn about serious diseases every day, are more prone to health anxiety and cyberchondria, an idea often called “medical students' disease”. Studies of medical students in the region point in different directions: most Egyptian medical students showed moderate cyberchondria [21], whereas in Palestine elevated health anxiety was least common among medical students [22].

## 1.8 Aim, objectives and hypotheses

**Aim:** To study the relationship between health anxiety and cyberchondria among adults in Baghdad who search for health information online, with a focus on AI chatbots and medical students.

**Objectives:**

1. To describe where adults in Baghdad look for health information (online and through AI) and what they do before seeing a doctor.
2. To estimate levels of health anxiety and cyberchondria and the strength of their association, independently of socio-demographic and digital-use factors (H1).
3. To compare health anxiety and cyberchondria between medical students and other participants (H2).

**Hypotheses:** (H1) People with more health anxiety have more cyberchondria, especially on the distress and excessiveness dimensions, independently of socio-demographic and digital-use factors. (H2) Medical students have higher health anxiety and cyberchondria than other participants.

H1 was the primary hypothesis and H2 secondary; all other analyses were exploratory (Figure 1).

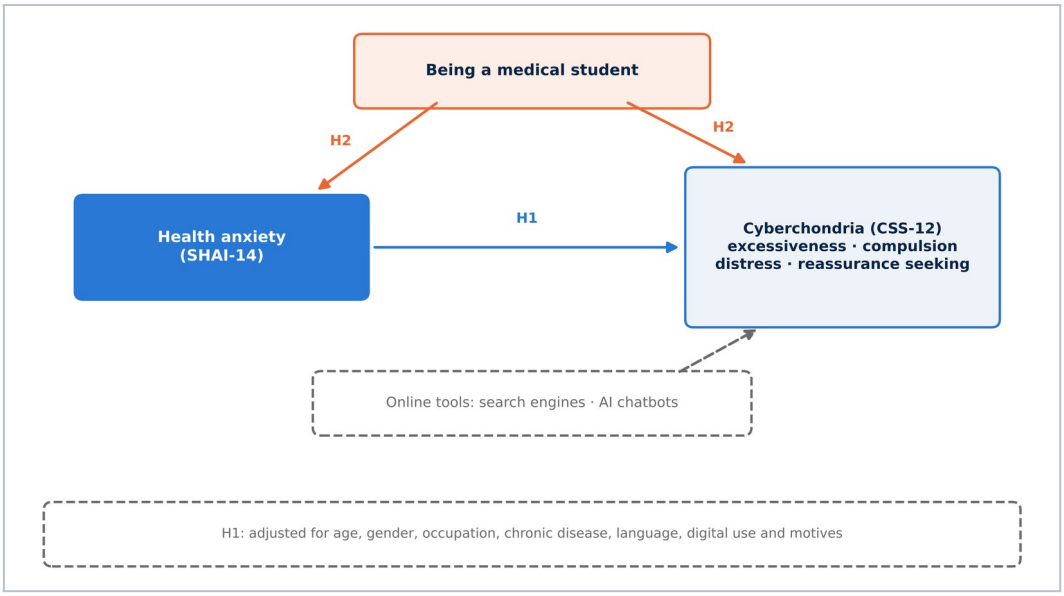



**Figure 1.** *Conceptual framework of the study: health anxiety is associated with cyberchondria independently of other factors (H1), and medical students are expected to have more health anxiety and cyberchondria (H2).*

## 2. Method

### 2.1 Design, setting and participants

This analytical cross-sectional study follows the STROBE [23] and CHERRIES [24] reporting guidelines. Data were collected online in Baghdad between 17 July 2026 and 11 September 2026 by convenience and snowball sampling through Baghdad-based social-media and messaging groups. Adults aged 18 or over who could read Arabic or English and gave informed consent were eligible. The form accepted only one response per Google account, and no duplicate response patterns were found; all completed submissions were analysed.

### 2.2 Sample size

Using Fisher's z with two-sided $\alpha = 0.05$ and 80% power, 194 participants are needed to detect a correlation of $r = 0.20$ [25]; meta-analytic

estimates are far larger ($r \approx 0.62$), so the target is conservative. For proportions, 300 participants give a margin of error of ±5.7%.

### 2.3 Questionnaire

A bilingual (Arabic–English) Google Form had five parts: (1) consent and socio-demographic data; (2) digital health-information seeking (tools, main AI chatbot, motives and trust in online/AI information); (3) the 14-item Short Health Anxiety Inventory (SHAI-14; items scored 0–3, total 0–42) [26], which has good reliability and validity [27,28]; (4) the CSS-12 (items scored 1–5, total 12–60; four 3-item subscales) [8]; and (5) behaviour after searching, disclosure of searches to the physician and the effect on the relationship with physicians. The Arabic wording was written by the team without formal cross-cultural adaptation [29]; published Arabic versions [9,30] were not used. The response options of one SHAI item (item 2) were displayed in a different order from the original and were scored by meaning.

### 2.4 Definitions

- **Elevated health anxiety:** SHAI-14 ≥ 18, the cut-off that best separated people with health anxiety from healthy controls in a Swedish diagnostic study [28]; it is a screening threshold, not a diagnosis, so prevalence at the study's other two cut-offs (22 and 29) is also reported.
- **High cyberchondria:** CSS-12 > 33 (upper tertile), as the CSS-12 has no validated cut-off.
- **AI-chatbot user:** ticked "AI chatbots" or named a chatbot; because 32 respondents named one without ticking it, all AI analyses were repeated with the strict (ticked-only) definition.
- **Health-related background:** medical students, other health-science students and healthcare workers.

- **Recoding for analysis:** the three oldest age groups (40–49, 50–59 and ≥ 60 years; n = 36) were merged into one group (≥ 40) because each was small, and free-text "Other" answers to the education and occupation questions were assigned to the closest existing category (for example, "PhD" to postgraduate, a secondary-school student to non-health student, and an engineer or homemaker to non-health employed/other).

### 2.5 Statistical analysis

Data were analysed with IBM SPSS Statistics version 26 (IBM Corp., Armonk, NY); two-sided $p < 0.05$ was considered significant. Categorical variables are presented as frequencies and percentages, with 95% confidence intervals for prevalence, and continuous variables as mean ± SD and median (IQR). Internal consistency was assessed with Cronbach's α. Because scale scores were not normally distributed (Shapiro–Wilk test), scores were compared between two groups with the Mann–Whitney U test and between three or more groups with the Kruskal–Wallis H test. Associations between categorical variables were tested with Pearson's $\chi^2$ test or Fisher's exact test, with Cramér's V as the effect size. The association between SHAI-14 and CSS-12 scores was measured with Pearson's (95% CI by Fisher's z transformation) and Spearman's correlation coefficients and with partial correlation controlling for covariates; Spearman's ρ was used to confirm the Pearson correlations because the scores were not normally distributed.

Independent predictors of the CSS-12 score were identified by hierarchical multiple linear regression (enter method) with socio-demographic variables in block 1, digital-use variables in block 2 and the SHAI-14 score in block 3; multicollinearity was checked with the variance inflation factor, and the normality of the residuals with a histogram and a normal P-P plot. Binary logistic regression (enter method) was used for elevated health anxiety, with results given as adjusted odds ratios (AOR) and 95% CI and model fit

assessed with the Hosmer–Lemeshow test and the area under the ROC curve. Medical students were compared with all other participants with the Mann–Whitney U and $\chi^2$ tests, and then with multiple linear regression (SHAI-14 and CSS-12 scores) and binary logistic regression (elevated health anxiety and high cyberchondria) adjusted for age group, gender, chronic disease and language (H2). As a sensitivity check, the main correlation was repeated after excluding straight-line responses and without SHAI item 2.

### 2.6 Ethical considerations

The study was an anonymous online survey of consenting adults, carried out outside any hospital or health facility, with no intervention and no access to medical records; it was conducted under the supervision of the College of Medicine, University of Baghdad. Participation was voluntary: the first page of the form explained the purpose of the study, electronic informed consent (with confirmation of age ≥ 18) was required before any question, and participants could withdraw at any time by closing the form. No names, e-mail addresses or contact details were collected, and the data were used only for this research.

## 3. Results

### 3.1 Participants and online health-information seeking

All 300 participants completed the questionnaire with no missing data. The sample was young (62.3% aged 20–29), 56.3% were female, 44.0% were medical students and 63.3% had a health-related background (Table 1). AI chatbots were the most used tool (78.0% ticked; 88.7% including those who named one), ahead of search engines (56.3%); 41.0% used AI without any search engine and ChatGPT was used by 59.3%. AI use fell from 96% at 18–19 years to 72% at 40 or older. The leading motives were convenience,

deciding whether a doctor was needed and instant reassurance (Table 2, Figure 2).

**Table 1.** Socio-demographic characteristics of the participants (n = 300)

| Characteristic | n | % |
|---|---|---|
| **Age group (years)** | | |
| 18-19 | 56 | 18.7 |
| 20-29 | 187 | 62.3 |
| 30-39 | 21 | 7.0 |
| 40-49 | 25 | 8.3 |
| 50-59 | 9 | 3.0 |
| 60+ | 2 | 0.7 |
| **Gender** | | |
| Female | 169 | 56.3 |
| Male | 131 | 43.7 |
| **Highest education** | | |
| Currently a student | 189 | 63.0 |
| High school or below | 18 | 6.0 |
| Diploma/Vocational | 7 | 2.3 |
| Bachelor | 65 | 21.7 |
| Postgraduate | 21 | 7.0 |
| **Study/work** | | |
| Medical student | 132 | 44.0 |
| Other health-science student | 43 | 14.3 |
| Healthcare worker | 15 | 5.0 |
| Non-health student | 46 | 15.3 |
| Non-health employee/other | 64 | 21.3 |
| **Health-related background (1=yes)** | | |
| Yes | 190 | 63.3 |
| No | 110 | 36.7 |
| **Diagnosed chronic condition (1=yes)** | | |
| Yes | 65 | 21.7 |
| No | 235 | 78.3 |
| **Language relied on** | | |
| Arabic | 212 | 70.7 |
| English | 88 | 29.3 |

**Table 2.** Online health information seeking, pre-consultation behaviour and relationship with physicians (n = 300)

| Variable | n | % |
|---|---|---|
| **Tools used (multiple)** | | |
| AI chatbots* | 266 | 88.7 |
| Search engines | 169 | 56.3 |
| Dedicated medical platforms | 55 | 18.3 |
| Social media | 32 | 10.7 |
| **Main AI chatbot** | | |
| ChatGPT | 178 | 59.3 |
| Gemini | 71 | 23.7 |
| Claude | 12 | 4.0 |
| Other AI | 3 | 1.0 |
| None reported | 36 | 12.0 |
| **Tool profile** | | |
| AI + search engine | 143 | 47.7 |
| AI without search engine | 123 | 41.0 |
| No AI (search/other only) | 34 | 11.3 |
| **Motivation (multiple)** | | |
| Convenience/accessibility | 240 | 80.0 |
| Check if doctor visit is needed | 173 | 57.7 |
| Instant reassurance | 153 | 51.0 |
| Self-diagnose / double-check prescription | 69 | 23.0 |
| Cost/logistic barriers | 54 | 18.0 |
| **Trust in online/AI medical data (1-5)** | | |
| 1 – Do not trust at all | 20 | 6.7 |
| 2 – Slightly | 83 | 27.7 |
| 3 – Moderately | 170 | 56.7 |
| 4 – Very much | 25 | 8.3 |
| 5 – Completely | 2 | 0.7 |
| **Behaviour after searching (multiple)** | | |
| Delay/skip care (reassured) | 104 | 34.7 |
| Ignore symptom | 90 | 30.0 |
| Visit multiple doctors | 76 | 25.3 |
| Urgent visit/ER/fast diagnosis | 66 | 22.0 |
| Self-medication | 64 | 21.3 |
| Delay care (terrified/overwhelmed) | 26 | 8.7 |
| **Disclose search to physician** | | |
| Always | 81 | 27.0 |
| Sometimes | 124 | 41.3 |
| Never (fear of judgement) | 95 | 31.7 |
| **Effect on doctor relationship** | | |
| No negative impact | 196 | 65.3 |
| Somewhat complicated | 66 | 22.0 |
| Significantly reduced trust | 38 | 12.7 |

*Note.* Tools, motivations and behaviours allowed multiple answers, so percentages do not sum to 100. AI users include 32 respondents who named a chatbot without ticking the AI option.

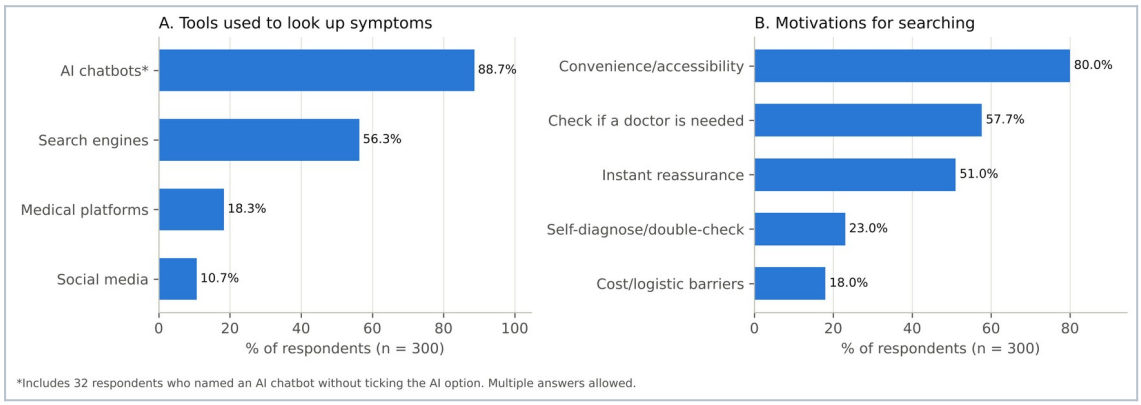



**Figure 2.** *Tools used and motivations for searching for health information online (n = 300).*

After searching, 34.7% delayed or skipped care because the information was reassuring, 30.0% ignored the symptom, 25.3% consulted several doctors and 22.0% sought urgent care. Only 27.0% always told their physician about their searches, whereas 31.7% never did, citing fear of judgement, and 34.7% said searching had complicated their relationship with, or trust in, their doctor (Figure 3).

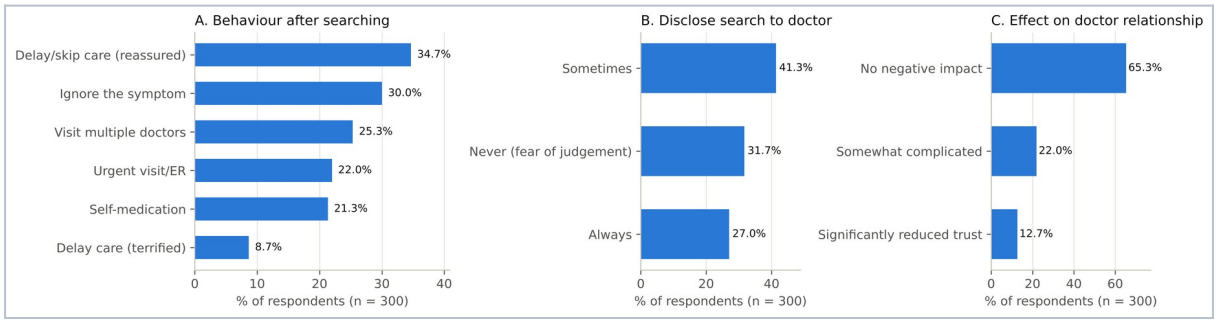



**Figure 3.** *Behaviour after searching, disclosure of searches to physicians and perceived effect on the doctor–patient relationship (n = 300).*

## 3.2 Levels of health anxiety and cyberchondria

Scale scores and reliability are shown in Table 3. The mean SHAI-14 score was 12.35 ± 5.83; 57 (19.0%) (95% CI 15.0–23.8%) had elevated health anxiety, an estimate that fell to 7.3% at a cut-off of 22 and 0.7% at 29 (Table 4). The mean CSS-12 score was 28.86 ± 8.36; participants searched frequently (excessiveness was the highest dimension) but searching rarely interfered with daily life (compulsion was the lowest). Both scales had good

internal consistency (Cronbach's α = 0.81 for the SHAI-14 and 0.86 for the CSS-12), and scores did not differ between participants who answered in Arabic and in English.

**Table 3.** Descriptive statistics and reliability of the SHAI-14 and CSS-12 (n = 300)

| Scale | Mean ± SD | Median (IQR) | Range | Cronbach's α |
|---|---|---|---|---|
| SHAI-14 total (0–42) | 12.35 ± 5.83 | 12 (8–16) | 1–30 | 0.81 |
| CSS-12 total (12–60) | 28.86 ± 8.36 | 29 (23–34) | 12–57 | 0.86 |
| Excessiveness (3–15) | 9.66 ± 2.94 | 10 (8–12) | 3–15 | 0.77 |
| Compulsion (3–15) | 5.00 ± 2.58 | 4 (3–6) | 3–15 | 0.88 |
| Distress (3–15) | 7.24 ± 2.74 | 7 (5–9) | 3–15 | 0.76 |
| Reassurance (3–15) | 6.96 ± 2.81 | 7 (5–9) | 3–15 | 0.76 |

*Note.* Cronbach's α ≥ 0.70 indicates acceptable internal consistency. Higher scores indicate more health anxiety (SHAI-14) and more cyberchondria (CSS-12).

**Table 4.** Prevalence of elevated health anxiety at different SHAI-14 cut-offs (n = 300)

| SHAI-14 score | Basis of the cut-off | n | % | 95% CI |
|---|---|---|---|---|
| **≥ 18** | **Optimal versus healthy controls (primary definition)** | **57** | **19.0** | **15.0–23.8** |
| ≥ 22 | Optimal in a psychiatric setting | 22 | 7.3 | 4.9–10.9 |
| ≥ 29 | Optimal in a low-prevalence setting | 2 | 0.7 | 0.2–2.4 |

*Note.* Cut-offs of 18, 22 and 29 from the Swedish diagnostic-accuracy study of the SHAI-14 [28]; ≥ 18 is used throughout this thesis. 95% CI: Wilson score interval. An elevated score is a screening result, not a diagnosis.

## 3.3 Association between health anxiety and cyberchondria

Health anxiety and cyberchondria were moderately correlated (r = 0.47, 95% CI 0.38–0.55; Spearman ρ = 0.47; p < 0.001), and the correlation was unchanged after adjustment for age, gender, health background, chronic disease, trust and AI use (partial r = 0.48) (Table 5, Figure 4). Health anxiety correlated with all four dimensions, most strongly with excessiveness (r = 0.46) and distress (r = 0.45) and least with compulsion (r = 0.25) and reassurance seeking (r = 0.24).

**Table 5.** Correlation between health anxiety (SHAI-14) and cyberchondria (CSS-12) and its dimensions (n = 300)

| Cyberchondria measure | Pearson r (95% CI) | Spearman ρ | p | Adjusted β |
|---|---|---|---|---|
| CSS-12 total | 0.47 (0.38–0.55) | 0.47 | <0.001 | 0.48 |
| Excessiveness | 0.46 (0.37–0.55) | 0.45 | <0.001 | 0.44 |
| Compulsion | 0.25 (0.14–0.36) | 0.32 | <0.001 | 0.29 |
| Distress | 0.45 (0.36–0.54) | 0.44 | <0.001 | 0.44 |
| Reassurance | 0.24 (0.13–0.35) | 0.23 | <0.001 | 0.25 |

*Note.* Pearson r with 95% CI calculated by Fisher's z transformation; all p < 0.001. Adjusted β: standardised coefficient of SHAI from multiple linear regression models adjusted for age, gender, occupation, chronic disease, language, tools, trust and motives.

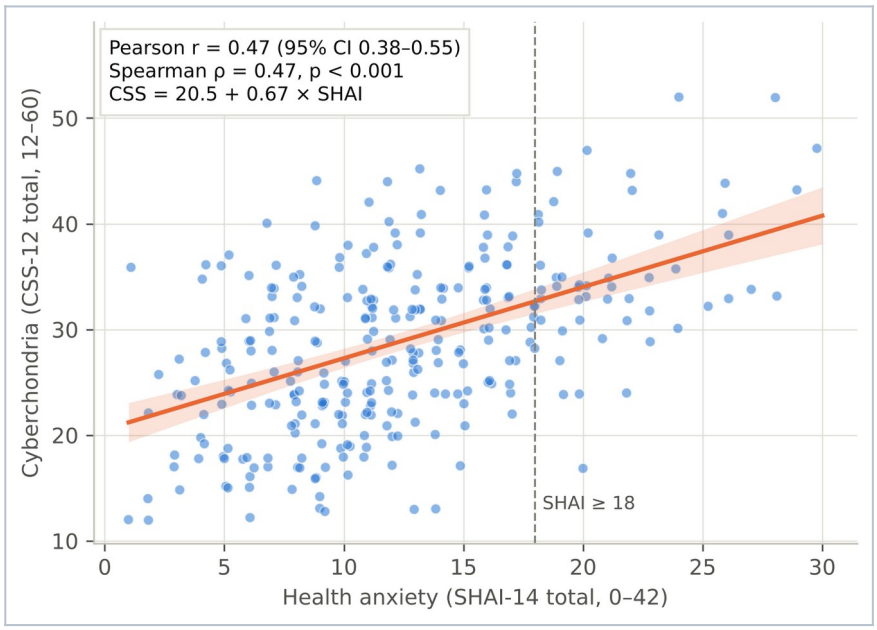



**Figure 4.** *Health anxiety (SHAI-14) against cyberchondria (CSS-12) with the least-squares line and 95% confidence band (n = 300).*

## 3.4 Factors associated with health anxiety and cyberchondria

Health anxiety (Table 6) was higher in participants with a chronic condition (p = 0.004), in those searching for instant reassurance (13.76 ± 5.65 vs 10.88 ± 5.65; p < 0.001), in social-media users (p = 0.001) and AI users (p = 0.008), and differed by age (p = 0.008), education (p = 0.002) and occupation (p = 0.005); gender and language made no difference. Cyberchondria was higher in AI users than in non-users (29.49 ± 8.21 vs 23.97 ± 8.05; p < 0.001), highest in those combining AI with search engines (p < 0.001; Figure 5), and was also higher in social-media users (p = 0.021).

Participants with elevated health anxiety more often had high cyberchondria (52.6% vs 23.0%; OR = 3.71, 95% CI 2.04–6.76) (Table 7).

**Table 6.** Health anxiety (SHAI-14) and cyberchondria (CSS-12) scores by participant characteristics (n = 300)

| Variable | n | SHAI mean ± SD | p | CSS mean ± SD | p |
|---|---|---|---|---|---|
| **Age group** | | | **0.008** | | **0.288** |
| 18-19 | 56 | 14.0 ± 5.5 | | 28.6 ± 8.1 | |
| 20-29 | 187 | 12.4 ± 5.8 | | 28.4 ± 8.0 | |
| 30-39 | 21 | 10.2 ± 6.4 | | 32.5 ± 11.3 | |
| ≥40 | 36 | 11.0 ± 5.7 | | 29.7 ± 8.2 | |
| **Gender** | | | **0.378** | | **0.859** |
| Female | 169 | 12.7 ± 6.0 | | 29.0 ± 8.7 | |
| Male | 131 | 11.9 ± 5.5 | | 28.6 ± 8.0 | |
| **Education** | | | **0.002** | | **0.779** |
| Currently a student | 189 | 13.3 ± 5.8 | | 28.6 ± 8.1 | |
| High school or below | 18 | 12.1 ± 5.4 | | 26.9 ± 6.6 | |
| Diploma/Vocational | 7 | 9.3 ± 2.7 | | 29.1 ± 8.3 | |
| Bachelor | 65 | 10.9 ± 5.9 | | 29.7 ± 8.8 | |
| Postgraduate | 21 | 9.7 ± 5.3 | | 30.0 ± 10.7 | |
| **Study/work** | | | **0.005** | | **0.820** |
| Medical student | 132 | 13.3 ± 5.8 | | 28.9 ± 8.1 | |
| Other health-science student | 43 | 12.2 ± 5.5 | | 29.2 ± 9.1 | |
| Healthcare worker | 15 | 10.4 ± 5.1 | | 29.5 ± 7.4 | |
| Non-health student | 46 | 13.0 ± 6.0 | | 27.6 ± 7.8 | |
| Non-health employee/other | 64 | 10.5 ± 5.7 | | 29.4 ± 9.1 | |
| **Chronic condition** | | | **0.004** | | **0.435** |
| Yes | 65 | 14.2 ± 6.0 | | 29.9 ± 8.9 | |
| No | 235 | 11.9 ± 5.7 | | 28.6 ± 8.2 | |
| **Uses AI chatbots** | | | **0.008** | | **<0.001** |
| Yes | 266 | 12.6 ± 5.7 | | 29.5 ± 8.2 | |
| No | 34 | 10.5 ± 6.7 | | 24.0 ± 8.1 | |
| **Uses search engines** | | | **0.111** | | **0.113** |
| Yes | 169 | 12.8 ± 5.9 | | 29.4 ± 7.9 | |
| No | 131 | 11.8 ± 5.7 | | 28.2 ± 8.9 | |
| **Uses social media** | | | **0.001** | | **0.021** |
| Yes | 32 | 15.8 ± 6.9 | | 32.5 ± 9.1 | |
| No | 268 | 11.9 ± 5.6 | | 28.4 ± 8.2 | |
| **Tool profile** | | | **0.006** | | **<0.001** |
| AI + search engine | 143 | 13.1 ± 5.6 | | 30.1 ± 7.6 | |
| AI without search engine | 123 | 12.0 ± 5.8 | | 28.7 ± 8.8 | |
| No AI (search/other only) | 34 | 10.5 ± 6.7 | | 24.0 ± 8.1 | |
| **Motive: reassurance** | | | **<0.001** | | **0.085** |
| Yes | 153 | 13.8 ± 5.7 | | 29.6 ± 7.6 | |
| No | 147 | 10.9 ± 5.7 | | 28.1 ± 9.1 | |
| **Motive: self-diagnosis** | | | **0.217** | | **0.079** |
| Yes | 69 | 13.1 ± 6.2 | | 30.4 ± 8.1 | |
| No | 231 | 12.1 ± 5.7 | | 28.4 ± 8.4 | |
| **Trust in online/AI** | | | **0.052** | | **0.110** |
| 1 | 20 | 10.1 ± 6.4 | | 24.8 ± 7.8 | |
| 2 | 83 | 13.2 ± 6.1 | | 28.9 ± 8.3 | |
| 3 | 170 | 11.9 ± 5.6 | | 28.9 ± 8.4 | |
| 4 | 25 | 14.7 ± 5.2 | | 31.8 ± 7.1 | |
| 5 | 2 | 13.0 ± 1.4 | | 25.5 ± 17.7 | |
| **Language** | | | **0.857** | | **0.654** |
| Arabic | 212 | 12.4 ± 5.7 | | 29.0 ± 8.2 | |
| English | 88 | 12.2 ± 6.1 | | 28.6 ± 8.8 | |

*Note.* p: Mann–Whitney U test (two groups) or Kruskal–Wallis H test (three or more groups); $p < 0.05$ was considered significant.

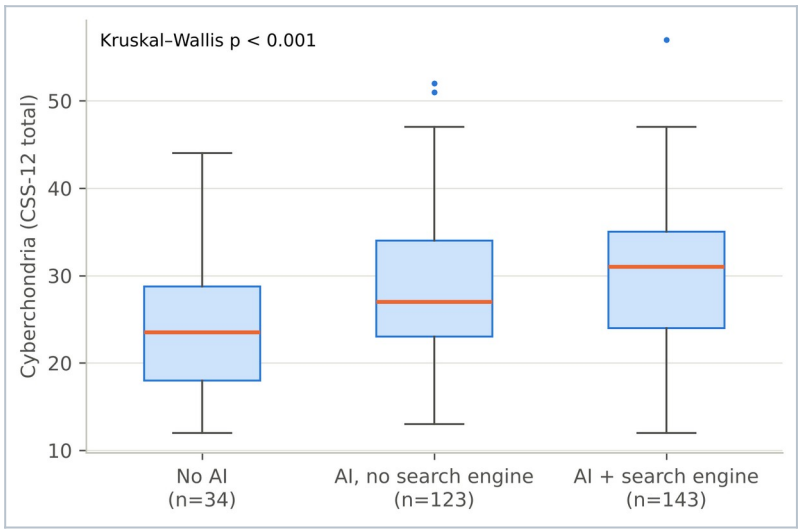



**Figure 5.** *Cyberchondria (CSS-12) by online tool profile.*

**Table 7.** Variables significantly associated (p < 0.05) with elevated health anxiety and with high cyberchondria

| Variable | n/N (%) with outcome, by group | Cramér's V | p |
|---|---|---|---|
| **Elevated health anxiety (SHAI ≥ 18)** | | | |
| Motive: reassurance | No: 15/147 (10.2%); Yes: 42/153 (27.5%) | 0.22 | <0.001 |
| Uses social media | No: 44/268 (16.4%); Yes: 13/32 (40.6%) | 0.19 | 0.001 |
| Delay care (terrified) | No: 47/274 (17.2%); Yes: 10/26 (38.5%) | 0.15 | 0.016 |
| Motive: convenience | No: 17/60 (28.3%); Yes: 40/240 (16.7%) | 0.12 | 0.039 |
| **High cyberchondria (CSS > 33)** | | | |
| Elevated health anxiety | No: 56/243 (23.0%); Yes: 30/57 (52.6%) | 0.26 | <0.001 |
| Disclose to physician | Always: 33/81 (40.7%); Never : 15/95 (15.8%); Sometimes: 38/124 (30.6%) | 0.21 | 0.001 |
| Ignore symptom | No: 70/210 (33.3%); Yes: 16/90 (17.8%) | 0.16 | 0.006 |
| Uses AI chatbots | No: 3/34 (8.8%); Yes: 83/266 (31.2%) | 0.16 | 0.007 |
| Tool profile | AI + search engine: 49/143 (34.3%); AI without search engine: 34/123 (27.6%); No AI (search/other only): 3/34 (8.8%) | 0.17 | 0.012 |
| Effect on doctor relationship | No negative impact: 46/196 (23.5%); Significantly reduced trust: 14/38 (36.8%); Somewhat complicated: 26/66 (39.4%) | 0.16 | 0.023 |
| Main AI chatbot | ChatGPT: 61/178 (34.3%); Claude: 4/12 (33.3%); Gemini: 17/71 (23.9%); None reported: 4/36 (11.1%); Other AI: 0/3 (0.0%) | 0.18 | 0.034 |
| Delay care (terrified) | No: 74/274 (27.0%); Yes: 12/26 (46.2%) | 0.12 | 0.039 |

*Note.* Pearson $\chi^2$ or Fisher's exact test (Monte-Carlo exact p where expected counts were < 5). Non-significant associations are available from the authors.

## 3.5 Independent predictors

In hierarchical regression (Table 8), socio-demographics explained 3% of the variance in cyberchondria (p = 0.53), digital use raised this to 13% (p <

0.001), and health anxiety added a further 18.4% ($R^2 = 0.317$, $p < 0.001$). Health anxiety was by far the strongest predictor (B = 0.68, $\beta = 0.48$, $p < 0.001$); participants aged 30–39 (B = 6.3) and 40 or older (B = 4.0) also had higher adjusted scores than those aged 20–29. AI use was associated with a 5.3-point higher CSS score under the inclusive definition ($p < 0.001$), but not under the strict definition (B = 1.49, $p = 0.210$), so there was no reliable evidence of an independent AI effect. In logistic regression, elevated health anxiety was independently associated with higher cyberchondria (AOR = 3.12 per SD, 95% CI 2.11–4.62), searching for instant reassurance (AOR = 3.83, 95% CI 1.80–8.18) and younger age (AOR = 0.61 per age-group step); the model fitted well (Hosmer–Lemeshow $p = 0.50$) and discriminated well (AUC = 0.83), although with about five events per predictor the odds ratios may be overestimated (Table 9, Figure 6). In the sensitivity analyses the correlation was unchanged ($r = 0.47$–$0.48$).

**Table 8.** Hierarchical multiple linear regression of cyberchondria (CSS-12 total) – final model (n = 300)

| Predictor | B | 95% CI | β | p |
|---|---|---|---|---|
| Constant | 12.04 | 6.66–17.42 | | <0.001 |
| Age 18–19 (ref. 20–29) | −0.43 | −2.85–2.00 | −0.02 | 0.730 |
| **Age 30–39** | **6.27** | **2.42–10.11** | **0.19** | **0.001** |
| **Age ≥40** | **3.98** | **0.16–7.80** | **0.15** | **0.041** |
| Healthcare worker (ref. non-health employed) | 1.06 | −3.27–5.39 | 0.03 | 0.630 |
| Medical student | 0.41 | −2.83–3.64 | 0.02 | 0.805 |
| Non-health student | −0.73 | −4.40–2.95 | −0.03 | 0.697 |
| Other health-science student | 1.32 | −2.31–4.95 | 0.06 | 0.474 |
| Female | −0.05 | −1.82–1.72 | 0.00 | 0.954 |
| Chronic condition | −0.14 | −2.20–1.92 | −0.01 | 0.895 |
| Answered mainly in English | 0.51 | −1.45–2.48 | 0.03 | 0.607 |
| **Uses AI chatbots** | **5.33** | **2.57–8.09** | **0.20** | **<0.001** |
| Uses search engines | 0.95 | −0.80–2.69 | 0.06 | 0.287 |
| Uses medical platforms | 0.17 | −2.14–2.48 | 0.01 | 0.886 |
| Uses social media | 1.46 | −1.31–4.24 | 0.05 | 0.299 |
| Trust in online/AI (per point) | 0.72 | −0.44–1.87 | 0.06 | 0.225 |
| Motive: instant reassurance | −1.11 | −2.91–0.70 | −0.07 | 0.228 |
| Motive: self-diagnosis | 0.99 | −1.10–3.09 | 0.05 | 0.352 |
| Motive: cost/logistics | 0.44 | −1.79–2.67 | 0.02 | 0.697 |
| Motive: need for a doctor? | 0.09 | −1.64–1.83 | 0.01 | 0.916 |
| **Health anxiety (SHAI, per point)** | **0.68** | **0.53–0.84** | **0.48** | **<0.001** |

*Note.* Block 1 (socio-demographics) $R^2 = 0.030$; Block 2 (+ digital use) $R^2 = 0.133$, $\Delta R^2 = 0.103$, $p = 0.0002$; Block 3 (+ SHAI) $R^2 = 0.317$, $\Delta R^2 = 0.184$, $p < 0.001$; adjusted $R^2 = 0.268$; $F = 6.48$, $p < 0.001$. β: standardised coefficient.

**Table 9.** Multivariable logistic regression of elevated health anxiety (SHAI ≥ 18; 57 events, n = 300)

| Predictor | AOR | 95% CI | p |
|---|---|---|---|
| **Age group (per step)** | **0.61** | **0.39–0.93** | **0.023** |
| Female | 1.23 | 0.62–2.43 | 0.558 |
| Health-related background | 1.05 | 0.48–2.27 | 0.905 |
| Chronic condition | 1.51 | 0.69–3.30 | 0.305 |
| Answered mainly in English | 0.94 | 0.45–1.96 | 0.870 |
| Uses AI chatbots | 0.61 | 0.18–2.13 | 0.442 |
| Uses search engines | 1.43 | 0.69–3.00 | 0.338 |
| Trust in online/AI (per point) | 0.74 | 0.46–1.19 | 0.217 |
| **Motive: instant reassurance** | **3.83** | **1.80–8.18** | **<0.001** |
| Motive: self-diagnosis | 0.93 | 0.43–2.03 | 0.863 |
| **Cyberchondria (per 1 SD)** | **3.12** | **2.11–4.62** | **<0.001** |

*Note.* Enter method. Omnibus $\chi^2(11) = 64.9$, $p < 0.001$; Nagelkerke $R^2 = 0.31$; Hosmer–Lemeshow $\chi^2(8) = 7.39$, $p = 0.50$; 81% correctly classified; area under the ROC curve = 0.83. Cyberchondria entered as a z-score.

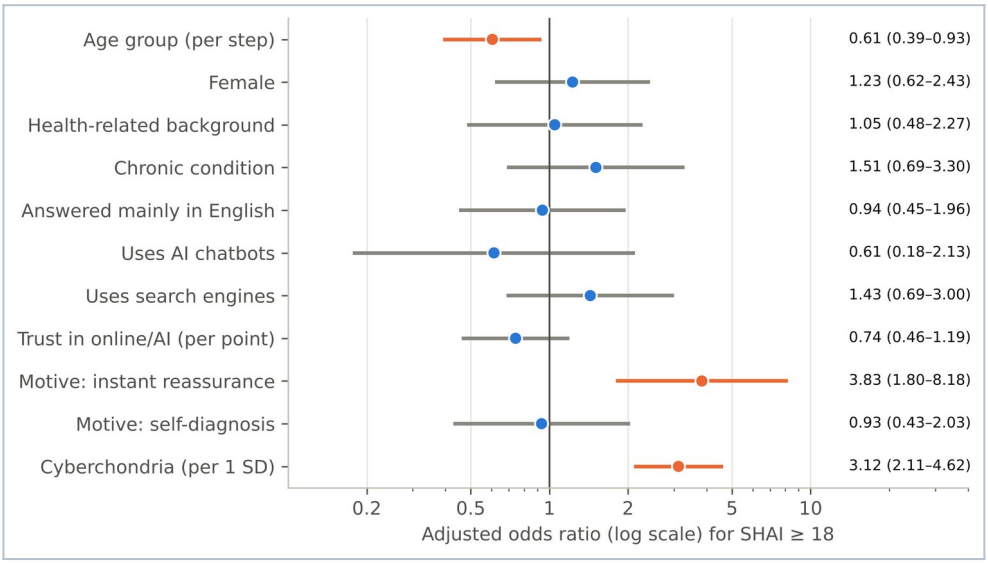



**Figure 6.** *Adjusted odds ratios for elevated health anxiety (SHAI ≥ 18); orange: p < 0.05.*

## 3.6 Medical students compared with other participants

Before adjustment, medical students had higher health anxiety than the other participants (13.3 ± 5.8 vs 11.6 ± 5.7; Mann–Whitney p = 0.004), and elevated health anxiety was somewhat more common among them (23.5% vs 15.5%; p = 0.079) (Table 10). After adjustment for age group, gender, chronic disease and language, the difference in health anxiety was no longer significant (B = 1.23, 95% CI −0.15–2.60, p = 0.081; elevated health anxiety AOR = 1.44, p = 0.238), because medical students were younger and younger participants had higher health anxiety. Cyberchondria was the same in medical students and the others (28.9 vs 28.9; p = 0.989), and remained so after adjustment. H2 was therefore not supported.

**Table 10.** Health anxiety and cyberchondria in medical students (n = 132) and other participants (n = 168)

| Measure | Medical students | Others | p | Adjusted effect (95% CI) | Adj. p |
|---|---|---|---|---|---|
| SHAI-14 score, mean ± SD | 13.3 ± 5.8 | 11.6 ± 5.7 | 0.004 | B 1.23 (−0.15–2.60) | 0.081 |
| Elevated health anxiety (SHAI ≥ 18), n (%) | 31 (23.5) | 26 (15.5) | 0.079 | AOR 1.44 (0.79–2.65) | 0.238 |
| CSS-12 score, mean ± SD | 28.9 ± 8.1 | 28.9 ± 8.6 | 0.989 | B 0.50 (−1.54–2.54) | 0.627 |
| High cyberchondria (CSS > 33), n (%) | 37 (28.0) | 49 (29.2) | 0.829 | AOR 1.08 (0.63–1.87) | 0.771 |

*Note.* p: Mann–Whitney U (scores) or $\chi^2$ (categories). Adjusted effect: B (linear regression) or AOR (logistic regression) for medical students vs all others, adjusted for age group, gender, chronic disease and language.

# 4. Discussion

## 4.1 Health anxiety and cyberchondria

To our knowledge, this is the first study to use the CSS-12 in Baghdad and one of the first to examine AI chatbots alongside search engines before a medical visit. Health anxiety and cyberchondria were moderately associated ($r = 0.47$), independently of the covariates, supporting H1. The association was concentrated in excessiveness and distress, the dimensions closest to fearful searching and its emotional aftermath, as the cognitive model predicts [5]. The 19% with elevated health anxiety is a screening estimate that varied from 0.7% to 19% across the three published cut-offs (18, 22 and 29), but it suggests that a substantial minority of digitally active adults in Baghdad report elevated health worry.

## 4.2 Comparison with worldwide and regional results

**Worldwide.** Our correlation was weaker than the meta-analytic estimates of 0.62–0.63 [12,13]. Low and narrowly spread health anxiety (mean SHAI 12.3), recruitment of confident Internet users and the short CSS-12, where most earlier studies used the 33-item scale, may all have weakened it. AI use here (78–88.7%) was far higher than in a 2024 US survey (about one in five) [16] or a 2025 survey (21%, against 98% for search engines) [31], partly because the sample was young.

**Iraq and the Arab region.** During the COVID-19 restrictions, 49.5% of Iraqi respondents screened positive for health anxiety [7]; that survey used a different questionnaire during a pandemic, so it is not directly comparable with our 19%, but both point to a substantial minority with elevated health worry. Our results agree with Arab-region data from Jeddah, Saudi primary care, Lebanon and 1435 Egyptian medical students, of whom 85.8% showed

moderate cyberchondria [9,18,19,21], and across 13 Arabic-speaking countries trust in online information predicted delayed consultation [20].

### 4.3 AI chatbots

The higher cyberchondria of AI users vanished under the strict definition and rested on a small, older comparison group: an absence of evidence, not evidence of no effect. Most participants trusted online and AI information only moderately, but trusted AI answers can still offer a route to false reassurance.

### 4.4 Medical students

H2 was not supported. Medical students reported more health anxiety than other participants only before adjustment; once age was taken into account the difference was no longer significant, because medical students were younger and younger participants were more anxious. Their cyberchondria was identical to that of the others. This argues against a strong “medical students' disease” in our sample: knowing more about disease did not make students search more anxiously, but it did not protect them either. The findings agree with a Palestinian study using the same SHAI threshold, in which elevated health anxiety was least common among medical students [22], and differ from a large Egyptian study in which most medical students had moderate cyberchondria [21], perhaps because that study used a different instrument and had no comparison group.

### 4.5 Searching and the consultation

Delaying care because of reassurance (34.7%) may save visits for minor symptoms but risks late diagnosis for serious ones. Almost a third never told their doctor about their searches, mainly for fear of being judged; information discussed openly strengthens the relationship, whereas hidden information undermines it [17]. Because these behaviours were described rather than

tested against health anxiety and cyberchondria, they show how common the problem is, not who is most affected.

### 4.6 Strengths and limitations

Strengths include two established, reliable scales, a bilingual questionnaire, separate assessment of AI chatbots, complete data, adjustment for confounders in multivariable models and sensitivity analyses. The main limitations are:

- **Design:** cross-sectional data cannot establish the direction of the association.
- **Sample:** an online convenience sample, young and with many medical students (44.0%), so percentages cannot be generalised to Baghdad; residence was not verified by a question.
- **Measurement:** Arabic wording without formal back-translation or a confirmatory factor analysis; one SHAI item with reordered options; CSS item 3 mentions only search engines and may miss repeated chatbot use; behaviour, trust and disclosure measured with single, non-validated items; all data self-reported.
- **Analysis:** many statistical tests were run without correction for multiple comparisons, so some significant results may be due to chance; small subgroups (34 AI non-users); about five events per predictor in the logistic model; the AI result depends on how 32 respondents were classified.

## 5. Conclusions and Recommendations

### 5.1 Conclusions

1. AI chatbots, mainly ChatGPT, are a leading source of health information before consulting a doctor among digitally active adults in Baghdad (78.0% ticked; 88.7% including those who named one).

2. About one in five (19%) had elevated health anxiety on a screening threshold; younger people and reassurance seekers were most affected.
3. Health anxiety and cyberchondria are moderately associated (r = 0.47), independently of other factors; this design cannot show which comes first.
4. Medical students did not have more health anxiety or cyberchondria than other participants once age was taken into account.
5. There is no reliable evidence that AI use adds to cyberchondria beyond health anxiety.
6. Searching shapes the consultation: about a third delay care when reassured and about a third never disclose their searches to their doctor.

### 5.2 Recommendations

- **Clinicians:** ask every patient, without judgement, what they have searched or asked an AI; treat repeated searching and reassurance seeking as possible signs of health anxiety; teach digital and AI health literacy in medical school.
- **Ministry of Health:** the Ministry may consider Arabic digital and AI health-literacy campaigns that warn about false reassurance, and reliable Arabic triage sources.
- **AI developers:** developers may consider anxiety-aware health answers that avoid escalating rare diagnoses and encourage appropriate consultation.
- **Research:** longitudinal studies, probability samples and formally adapted Arabic scales; intervention trials are needed, as ClinicalTrials.gov lists only nine cyberchondria studies and none in Iraq [32].

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# Appendix. Study questionnaire

*Administered online through Google Forms in English and Arabic (every item shown in both languages; the Arabic text is not reproduced here). All items were required except Q7 and Q9. ○ = one answer; ☐ = all that apply.*

**Part 1. Consent and socio-demographic data**

**1.** I confirm that I am at least 18 years of age, have read the description of this study and voluntarily agree to participate; I can withdraw at any point without penalty.
○ *I explicitly agree and consent to participate*

**2.** Age group
○ *18–19 · 20–29 · 30–39 · 40–49 · 50–59 · 60+*

**3.** Gender
○ *Male · Female*

**4.** Highest level of education completed
○ *Currently a student · Postgraduate (Master's or PhD) · Bachelor's · Diploma/vocational · High school or below · Other*

**5.** Nature of current study or work
○ *Medical student · Student in another healthcare field · Medical doctor · Other healthcare professional · Student in a non-healthcare field · Employed in a non-healthcare field · Other*

**6.** Do you have a diagnosed chronic medical condition?
○ *Yes · No*

**7.** If yes, please specify (optional, free text).

**Part 2. Digital health information seeking**

**8.** Which tools do you use most frequently to look up symptoms online?
☐ *Search engine (e.g. Google) · AI chatbots · Dedicated medical platforms (e.g. WebMD, Mayo Clinic) · Social media (e.g. Facebook, Reddit, Instagram)*

**9.** If you use AI, which one most of the time?
○ *ChatGPT · Gemini · Claude · Other*

**10.** Primary motivation for searching medical information online
☐ *Convenience and easy access · High cost or difficulty of doctor visits · Instant reassurance about an unexplained symptom · Self-diagnosis or double-checking a prescription · To see if a symptom is serious enough to need a doctor · Other*

**11.** How much do you trust medical information generated by online tools/AI (1–5)?
○ *1 Not at all · 2 Slightly · 3 Moderately · 4 Very much · 5 Completely*

**Part 3. Short Health Anxiety Inventory (SHAI-14): choose the statement that best describes you (scored 0–3)**

**12.** Worrying about health
*I do not worry about my health / occasionally worry / spend much of my time worrying / spend most of my time worrying*

**13.** Body awareness*
*I notice aches/pains as much as most people (of my age) / less than most people / more than most people / I am aware of aches/pains all the time*

**14.** Bodily sensations
*As a rule I am not aware of bodily sensations or changes / sometimes aware / often aware / constantly aware*

**15.** Thoughts of illness
*Resisting thoughts of illness is never a problem / most of the time I can resist them / I try but am often unable to / they are so strong I no longer try*

**16.** Fear of serious illness
*As a rule I am not afraid I have a serious illness / sometimes afraid / often afraid / always afraid*

**17.** Images of illness
*I do not have images of myself being ill / occasionally / frequently / constantly*

**18.** Distraction from health thoughts
*No difficulty taking my mind off my health / sometimes difficult / often difficult / nothing can take my mind off it*

**19.** Medical reassurance
*Lastingly relieved if my doctor says nothing is wrong / relieved but worries sometimes return / relieved but worries always return / not relieved*

**20.** Hearing about illness
*If I hear about an illness I never think I have it / sometimes / often / always*

**21.** Interpretation of sensations
*If I have a bodily sensation I rarely wonder what it means / often wonder / always wonder / must know what it means*

**22.** Perceived risk
*I usually feel at very low / fairly low / moderate / high risk of developing a serious illness*

**23.** Thinking you are ill
*I never / sometimes / often think I have a serious illness / I usually think I am seriously ill*

**24.** Unexplained sensations
*If I notice an unexplained sensation I do not find it difficult / sometimes / often / always find it difficult to think about other things*

**25.** Family and friends
*My family/friends would say I do not worry enough / have a normal attitude / worry too much about my health / am a hypochondriac*

*Options for Q13 were displayed in the order shown (the original scale lists "less than" before "as much as"); scoring followed the meaning of each statement.*

**Part 4. Cyberchondria Severity Scale (CSS-12): ○ Never (1) · Rarely (2) · Sometimes (3) · Often (4) · Always (5)**

**26.** If I notice an unexplained bodily sensation, I search for it on the Internet.

**27.** I read different web pages about the same health condition.

**28.** I enter the same symptoms into a search engine on more than one occasion.

**29.** Researching symptoms online distracts me from reading news, sports or entertainment.

**30.** Researching symptoms online interrupts my work (e.g. emails, documents).

**31.** Researching symptoms online interrupts my offline social activities.

**32.** I panic when I read online that a symptom I have could be a sign of a rare or serious condition.

**33.** I feel fine until I read about a serious medical condition online.

**34.** I feel more anxious or distressed after researching symptoms online.

**35.** Researching symptoms online leads me to consult my general practitioner.

**36.** I suggest to my doctor a specific diagnostic procedure I have read about online (e.g. a biopsy, a blood test).

**37.** Researching symptoms online leads me to consult other medical specialists.

**Part 5. Pre-consultation behaviour and the doctor–patient relationship**

**38.** After searching your symptoms online, what are your most frequent responses?
*☐ Buy self-prescribed medication without a doctor · Delay or skip medical help because the information reassured me · Delay help because the information frightened or overwhelmed me · Seek an urgent appointment or the emergency room · Visit several doctors to double-check · Ignore the symptom*

**39.** During a consultation, do you share your online or AI findings with your physician?
*○ Yes, always · Sometimes, depending on how the doctor responds · No, never, for fear of being judged or dismissed*

**40.** How has digital health information-seeking affected your relationship with and trust in healthcare providers?
*○ Significantly reduced my trust · Somewhat complicated the relationship, but I still follow advice · No negative impact; it supplements my doctor's care*

**41.** Which language did you rely on most while completing this questionnaire?
*○ Mostly English · Mostly Arabic*